1. Evidence review: the isogeny-chain encoding with the chain variables eliminated first (m = 3)

Decision DEC-20261019-b01514: **reject_scoped** (scope widened to block elimination orders at $m = 3$) · **evidence** EV-ECDLP-97cb57 (contradicts, replicated, toy tier) · **validation** VAL-20261019-e750db (passed) · **runs** RUN-ECDLP-20a588-01, −03 · **experiment** EXP-ECDLP-20a588 · **goal** GOAL-ECDLP-001.

Scoped claim. For Semaev $m = 3$ on toy primes (10–22 bits, $L = 3..7$), solving the chain encoding of the factor-base set with the chain variables eliminated first (msolve `-e`, Singular block order) removes the growing penalty of plain grevlex but stays slower than the dense encoding of the same set: msolve paired median ratios 1.34, 1.86, 2.27, 1.94 at $L = 3..6$ (95% intervals all above 1); Singular 1.61 and 1.42 at $L = 3, 4$; both msolve arms are retired together at $L = 7$.

1.1. Question

EXP-ECDLP-502ae2 left one route open: the chain's equations are triangular, so a block order that eliminates the chain variables first might undo the loss that plain DRL showed. This experiment tests that on the same $m = 3$ instances with both solvers.

1.2. Observation

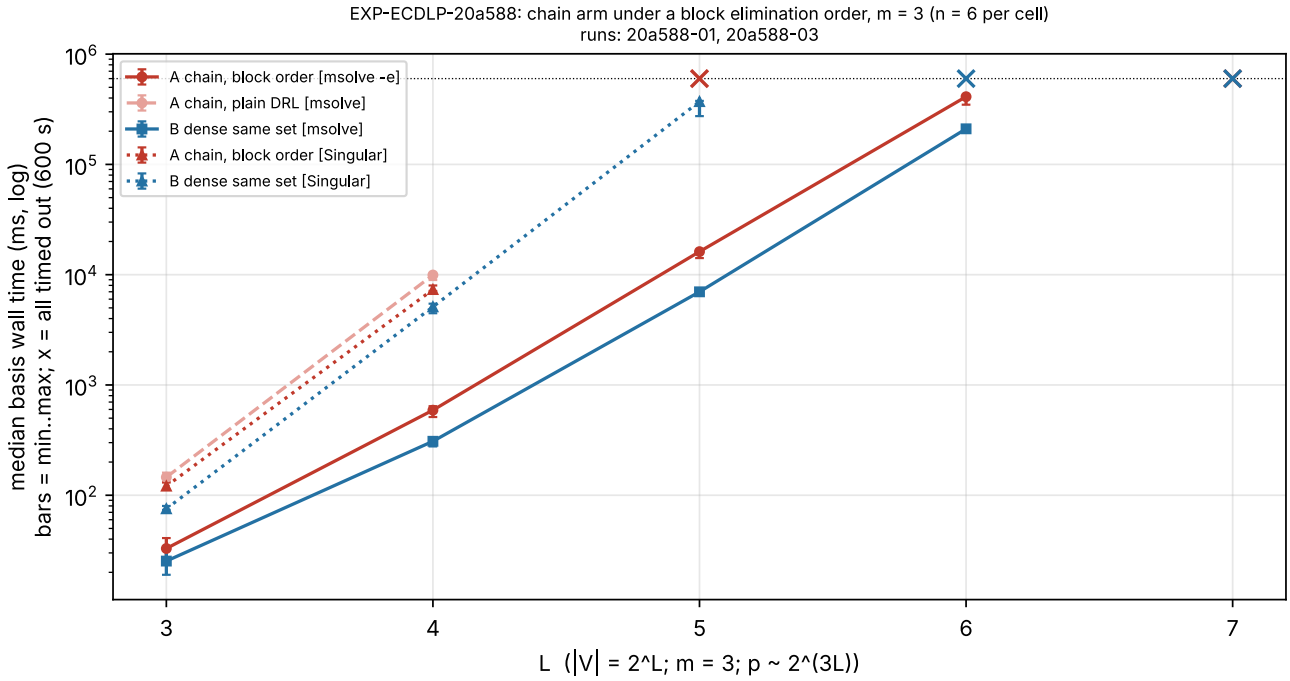


Figure 1: Median basis time per arm vs L (measured); bars min–max over completed instances; × = every instance timed out at 600 s; $n = 6$ per cell. Runs RUN-01 ($L = 3..6$) and RUN-03 ($L = 7$).

comparison	derived from the recorded medians (criteria.json)
msolve A_elim vs B	jointly completed L = [3, 4, 5, 6]; A_elim_ms faster in 0 of 4 cells; median ratio A_elim_ms/B_ms by L: L3: 1.3, L4: 1.91, L5: 2.31, L6: 1.96; largest L completed: A_elim_ms 6, B_ms 6; retired (every non-degenerate instance timed out) at L: A_elim_ms 7, B_ms 7
Singular A_block vs B	jointly completed L = [3, 4]; A_block_sg faster in 0 of 2 cells; median ratio A_block_sg/B_sg by L: L3: 1.61, L4: 1.43; largest L completed: A_block_sg 4, B_sg 5; retired (every non-degenerate instance timed out) at L: A_block_sg 5, B_sg 6
dynamic range A_drl vs A_elim (msolve)	jointly completed L = [3, 4]; A_drl_ms faster in 0 of 2 cells; median ratio A_drl_ms/A_elim_ms by L: L3: 4.43, L4: 16.81; largest L completed: A_drl_ms 4, A_elim_ms 6; retired (every non-degenerate instance timed out) at L: A_drl_ms 5, A_elim_ms 7
elim_equals_B	eliminated chain basis equals the dense basis: L3: 6/6, L4: 6/6, L5: 6/6, L6: 6/6

Paired per-instance ratios (supplementary, not a frozen criterion): the two arms run on the same instance; median of the ratios with a 95% percentile bootstrap interval (4000 resamples, deterministic seed). Intervals are wide for $n = 4..6$ and are reported as such.

pair	L	n	median ratio	95% CI
ms A elim over B	L3	6	1.34	1.11–1.58
ms A elim over B	L4	6	1.86	1.82–2.04
ms A elim over B	L5	6	2.27	2.18–2.35
ms A elim over B	L6	6	1.94	1.78–1.99
sg A block over B	L3	6	1.60	1.52–1.64
sg A block over B	L4	6	1.42	1.33–1.73
ms A drl over A elim	L3	6	4.57	3.71–4.95
ms A drl over A elim	L4	6	16.21	15.58–17.91

L	arm	ok/to/deg	median ms [min–max]	planted	max F4 deg
3	A_block_sg	6/0/0	121 [112–130]	4/4	—
3	A_drl_ms	6/0/0	146 [133–160]	4/4	12
3	A_elim_ms	6/0/0	33 [27–41]	4/4	14
3	B_ms	6/0/0	25 [19–32]	4/4	16
3	B_sg	6/0/0	75 [74–80]	4/4	—
4	A_block_sg	6/0/0	7,329 [7,116–7,984]	4/4	—
4	A_drl_ms	6/0/0	9,933 [8,947–10,140]	4/4	13
4	A_elim_ms	6/0/0	591 [512–642]	4/4	32
4	B_ms	6/0/0	309 [277–336]	4/4	24
4	B_sg	6/0/0	5,112 [4,472–5,458]	4/4	—
5	A_block_sg	0/6/0	— [——]	—	—
5	A_drl_ms	0/6/0	— [——]	—	—
5	A_elim_ms	6/0/0	16,183 [14,145–16,526]	4/4	65
5	B_ms	6/0/0	6,993 [6,559–7,195]	4/4	40
5	B_sg	6/0/0	370,938 [274,132–376,859]	4/4	—
6	A_elim_ms	6/0/0	411,159 [347,985–418,707]	4/4	133
6	B_ms	6/0/0	210,271 [194,599–213,522]	4/4	72
6	B_sg	0/6/0	— [——]	—	—
7	A_elim_ms	0/6/0	— [——]	—	—
7	B_ms	0/6/0	— [——]	—	—

1.3. Comparison

- **The order matters a lot, then stops mattering:** plain grevlex is 4.6 × slower than the block order at $L = 3$ and 16 × slower at $L = 4$ (the dynamic-range control passes). The block-order chain is then about 2 × the dense arm, flat in L within the intervals (1.34, 1.86, 2.27, 1.94), not growing like the plain-DRL gap (8× → >79×).
- **Same ideal:** the eliminated chain basis equals the dense basis on all 24 instances where both finished, so the extra cost is the chain rows and the path, not a different problem.
- **Singular agrees in sign:** block-order chain 1.6× and 1.4× at $L = 3, 4$, retired at $L = 5$ against the dense arm's $L = 6$.
- **Exploratory (not preregistered):** msolve's maximum F4 degree on the dense arm is $2^L + 8$ on every instance; on the chain arm with block order it is 14, 32, 65, 133 for instances that have a solution (about 2^{L+1}) and close to the dense value on instances with no solution (7 of 8 of those have the unit ideal as basis, which ends the computation early). Treat as a lead, not a result.

1.4. Inference

A block order recovers the chain's plain-DRL loss but leaves a roughly constant factor of about 2 over the dense encoding for $L = 3..6$ under both solvers; it does not make the chain competitive, and the retirement comparison at $L = 7$ is a tie of right-censored cells. The L-independent-degree route is not supported (the degree observation points the other way).

1.5. Limitations

- Toy scale: $m = 3$, 10–22-bit primes; $n = 6$ per cell (percentile bootstrap at $n = 6$ is coarse). Only one block-order configuration per solver; other weighted or interleaved orders untested.
- All twelve $L = 7$ solves are right-censored at 600 s: they set the retirement L under the frozen stopping rule and say nothing about relative cost. The ratio's trend beyond $L = 6$ is unmeasured; the validator's probes show memory growing about 17–19× per L (A: 2.7 GB, B: 1.3 GB at $L = 6$), so it may bind at $L = 7$.
- Process: RUN-01 was killed by the host 2-hour limit in $L = 7$; RUN-02 (the AMD-01 re-run) was lost to a VM recycle; RUN-03 (AMD-02) ran $L = 7$ for the two arms the stopping rule had not retired. No 17-bit prime was used, so msolve's crash on 17-bit primes (CORR-20261019-e6f65b) affects no value here. Resource records (peak memory, CPU time) were not kept in the run receipts.

1.6. Next actions

1. Account for F4 matrix dimensions (msolve `-v 2` logs) for chain vs dense per L, the first discriminating experiment in the reframed problem.
2. Read the $m = 4$ experiment (EXP-ECDLP-8c41d7) for the m-dependence of the constant factor.
3. No further P-256 work on this route unless some order or encoding beats the dense and random controls on the frozen grid.